

Gene

A Schema.org Type

This term is in the "new" area - implementation feedback and adoption from applications and websites can help improve our definitions.

Thing > **BioChemEntity** > **Gene**

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A discrete unit of inheritance which affects one or more biological traits (Source: <https://en.wikipedia.org/wiki/Gene>). Examples include FOXP2 (Forkhead box protein P2), SCARNA21 (small Cajal body-specific RNA 21), A- (agouti genotype).

Property	Expected Type	Description
Properties from Gene		
alternativeOf	Gene	Another gene which is a variation of this one.
encodesBioChemEntity	BioChemEntity	Another BioChemEntity encoded by this one. Inverse property: isEncodedByBioChemEntity
expressedIn	AnatomicalStructure or AnatomicalSystem or	Tissue, organ, biological sample, etc in which activity of this gene has been observed experimentally. For example brain, digestive system.

Property	Expected Type	Description
	BioChemEntity or DefinedTerm	
hasBioPolymerSequence	Text	A symbolic representation of a BioChemEntity. For example, a nucleotide sequence of a Gene or an amino acid sequence of a Protein.
Properties from BioChemEntity		
associatedDisease	MedicalCondition or PropertyValue or URL	Disease associated to this BioChemEntity. Such disease can be a MedicalCondition or a URL. If you want to add an evidence supporting the association, please use PropertyValue.
bioChemInteraction	BioChemEntity	A BioChemEntity that is known to interact with this item.
bioChemSimilarity	BioChemEntity	A similar BioChemEntity, e.g., obtained by fingerprint similarity algorithms.
biologicalRole	DefinedTerm	A role played by the BioChemEntity within a biological context.
funding	Grant	A Grant that directly or indirectly provide funding or sponsorship for this item. See also ownershipFundingInfo. Inverse property: fundedItem
hasBioChemEntityPart	BioChemEntity	Indicates a BioChemEntity that (in some sense) has this BioChemEntity as a part. Inverse property: isPartOfBioChemEntity
hasMolecularFunction	DefinedTerm or PropertyValue or URL	Molecular function performed by this BioChemEntity; please use PropertyValue if you want to include any evidence.

Property	Expected Type	Description
hasRepresentation	PropertyValue or Text or URL	A common representation such as a protein sequence or chemical structure for this entity. For images use schema.org/image.
isEncodedByBioChemEntity	Gene	Another BioChemEntity encoding by this one. Inverse property: encodesBioChemEntity
isInvolvedInBiologicalProcess	DefinedTerm or PropertyValue or URL	Biological process this BioChemEntity is involved in; please use PropertyValue if you want to include any evidence.
isLocatedInSubcellularLocation	DefinedTerm or PropertyValue or URL	Subcellular location where this BioChemEntity is located; please use PropertyValue if you want to include any evidence.
isPartOfBioChemEntity	BioChemEntity	Indicates a BioChemEntity that is (in some sense) a part of this BioChemEntity. Inverse property: hasBioChemEntityPart
taxonomicRange	DefinedTerm or Taxon or Text or URL	The taxonomic grouping of the organism that expresses, encodes, or in some way related to the BioChemEntity.
Properties from Thing		
additionalType	Text or URL	An additional type for the item, typically used for adding more specific types from external vocabularies in microdata syntax. This is a relationship between something and a class that the thing is in. Typically the value is a URI-identified RDF class, and in this case corresponds to the use of rdf:type in RDF. Text values can be used

Property	Expected Type	Description
		sparingly, for cases where useful information can be added without their being an appropriate schema to reference. In the case of text values, the class label should follow the schema.org style guide .
alternateName	Text	An alias for the item.
description	Text or TextObject	A description of the item.
disambiguatingDescription	Text	A sub property of description. A short description of the item used to disambiguate from other, similar items. Information from other properties (in particular, name) may be necessary for the description to be useful for disambiguation.
identifier	PropertyValue or Text or URL	The identifier property represents any kind of identifier for any kind of Thing , such as ISBNs, GTIN codes, UUIDs etc. Schema.org provides dedicated properties for representing many of these, either as textual strings or as URL (URI) links. See background notes for more details.
image	ImageObject or URL	An image of the item. This can be a URL or a fully described ImageObject .
mainEntityOfPage	CreativeWork or URL	Indicates a page (or other CreativeWork) for which this thing is the main entity being described. See background notes for details. Inverse property: mainEntity
name	Text	The name of the item.
potentialAction	Action	Indicates a potential Action, which describes an idealized action in which this thing would play an 'object' role.

Property	Expected Type	Description
sameAs	URL	URL of a reference Web page that unambiguously indicates the item's identity. E.g. the URL of the item's Wikipedia page, Wikidata entry, or official website.
subjectOf	CreativeWork or Event	A CreativeWork or Event about this Thing. Inverse property: about
url	URL	URL of the item.

Instances of **Gene** may appear as a value for the following properties

Property	On Types	Description
alternativeOf	Gene	Another gene which is a variation of this one.
isEncodedByBioChemEntity	BioChemEntity	Another BioChemEntity encoding by this one.

Source

<http://bioschemas.org>

